EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. Complete set of values of K in order that $f(x) = \sin x - \cos x - Kx + b$ decreases for all real values is given by -

(A) $K < 1$ (B) $K \geq 1$ (C) $K \geq \sqrt{2}$ (D) $K < \sqrt{2}$

MMN001

2. If the function $f(x) = 2x^2 - kx + 5$ is strictly increasing in $[1, 2]$, then complete set of values of k is

(A) $(-\infty, 4)$ (B) $(4, \infty)$ (C) $(-\infty, 8]$ (D) $(8, \infty)$

MMN002

3. If $f(x) = x^3 - 10x^2 + 200x - 10$, then $f(x)$ is-

(A) strictly decreasing in $(-\infty, 10]$ and strictly increasing in $[10, \infty)$

(B) strictly increasing in $(-\infty, 10]$ and strictly decreasing in $[10, \infty)$

(C) strictly increasing for every value of x

(D) strictly decreasing for every value of x

MMN004

4. The function $f(x) = \tan^{-1}(\sin x + \cos x)$ is an increasing function in -

(A) $(\pi/4, \pi/2)$ (B) $(-\pi/2, \pi/4)$ (C) $(0, \pi/2)$ (D) $(-\pi/2, \pi/2)$

MMN012

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5. A function is matched below against an interval where it is supposed to be increasing. which of the following pairs is incorrectly matched ?

	interval	**function**
(A)	$(-\infty, \infty)$	$x^3 - 3x^2 + 3x + 3$
(B)	$[2, \infty)$	$2x^3 - 3x^2 - 12x + 6$
(C)	$\left(-\infty, \frac{1}{3}\right]$	$3x^2 - 2x + 1$
(D)	$(-\infty, -4)$	$x^3 + 6x^2 + 6$

MMN015

6. If function $f(x) = 2x^2 + 3x - m\ell nx$ is monotonic decreasing in the interval $(0, 1)$, then the least value of the parameter m is -

(A) 7 (B) $\frac{15}{2}$ (C) $\frac{31}{4}$ (D) 8

MMN003

7. If $f: [1, 10] \to [1, 10]$ is a non-decreasing function and $g: [1, 10] \to [1, 10]$ is a non-increasing function. Let $h(x) = f(g(x))$ with $h(1) = 1$, then $h(2)$

(A) lies in $(1, 2)$ (B) is more than 2 (C) is equal to 1 (D) is not defined

MMN014

8. If $2a + 3b + 6c = 0$, then at least one root of the equation $ax^2 + bx + c = 0$ lies in the interval-

(A) $(0, 1)$ (B) $(1, 2)$ (C) $(2, 3)$ (D) none

MMN005

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9. If the function $f(x) = x^3 - 6x^2 + ax + b$ defined on [1, 3], satisfies the rolle's theorem for $c = \frac{2\sqrt{3}+1}{\sqrt{3}}$, then -

(A) $a = 11, b = 6$ (B) $a = -11, b = 6$ (C) $a = 11, b \in R$ (D) None of these

MMN007

10. The equation $\sin x + x \cos x = 0$ has at least one root in

(A) $\left(-\frac{\pi}{2}, 0\right)$ (B) $(0, \pi)$ (C) $\left(\pi, \frac{3\pi}{2}\right)$ (D) $\left(0, \frac{\pi}{2}\right)$

MMN011

11. A value of c for which the conclusion of Mean values theorem holds for the function $f(x) = \log_e x$ on the interval [1, 3] is -

(A) $2\log_3 e$ (B) $\frac{1}{2}\log_e 3$ (C) $\log_3 e$ (D) $\log_e 3$

MMN006

12. Given: $f(x) = 4 - \left(\frac{1}{2} - x\right)^{2/3}$; $g(x) = \begin{cases} \frac{\tan[x]}{x} & , x \neq 0 \\ 1 & , x = 0 \end{cases}$

$h(x) = \{x\}$; $k(x) = 5^{\log_2(x+3)}$

then in [0, 1], Lagranges Mean Value Theorem is NOT applicable to

(A) f, g, h (B) h, k (C) f, g (D) g, h, k

where $[x]$ and $\{x\}$ denotes the greatest integer and fractional part function.

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MMN008

13. Given $f'(1) = 1$ and $\frac{d}{dx}(f(2x)) = f'(x)\ \forall\ x > 0$. If $f'(x)$ is differentiable then there exists a number $c \in (2, 4)$ such that $f''(c)$ equals

(A) – 1/4 (B) – 1/8 (C) 1/4 (D) 1/8

MMN009

14. If the function $f(x) = 2x^2 + 3x + 5$ satisfies LMVT at $x = 2$ on the closed interval $[1, a]$, then the value of 'a' is equal to

(A) 3 (B) 4 (C) 6 (D) 1

MMN010

15. The function $f(x) = (x - 3)^2$ satisfies all the conditions of Lagrange's mean value theorem in [3,4]. A point on $y = (x - 3)^2$, where the tangent is parallel to the chord joining (3, 0) and (4, 1) is

(A) $\left(\frac{7}{2}, \frac{1}{2}\right)$ (B) $\left(\frac{7}{2}, \frac{1}{4}\right)$ (C) (1, 4) (D) (4, 1)

MMN013

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EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. Which of the following statements is/are correct?

(A) $x + \sin x$ is increasing function

(B) $\sec x$ is neither increasing nor decreasing function

(C) $x + \sin x$ is decreasing function

(D) $\sec x$ is an increasing function

MMN037

2. If $f(x) = 2x + \cot^{-1}x + \ell n\left(\sqrt{1+x^2} - x\right)$, then $f(x)$:

(A) increases in $[0, \infty)$ (B) decreases in $[0, \infty)$

(C) neither increases nor decreases in $[0, \infty)$ (D) increases in $(-\infty, \infty)$

MMN038

3. Let $f(x) = \sin^2 x$ & $0 < A < B$, where $A, B \in \left(0, \frac{\pi}{4}\right)$, then

(A) $(B - A)\sin 2A < \sin^2 B - \sin^2 A < (B - A)\sin 2B$

(B) $(B - A)\sin 2A > \sin^2 B - \sin^2 A > (B - A)\sin 2B$

(C) $\cos 2B < \cos 2A$

(D) $B\cos 2A > A\cos 2B$

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MMN042

4. Let $f(x)$ be a differentiable function and $f(\alpha) = f(\beta) = 0$ $(\alpha < \beta)$, then in the interval (α, β)

(A) $f(x) + f'(x) = 0$ has at least one root

(B) $f(x) - f'(x) = 0$ has at least one real root

(C) $f(x) \cdot f'(x) = 0$ has at least one real root

(D) none of these

MMN041

5. Let $\phi(x) = (f(x))^3 - 3(f(x))^2 + 4f(x) + 5x + 3\sin x + 4\cos x \ \forall\ x \in R$, where $f(x)$ is a differentiable function $\forall\ x \in R$, then

(A) ϕ is increasing whenever f is increasing

(B) ϕ is increasing whenever f is decreasing

(C) ϕ is decreasing whenever f is decreasing

(D) ϕ is decreasing if $f'(x) = -11$

MMN040

6. Let $g(x) = 2f(x/2) + f(1 - x)$ and $f''(x) < 0$ in $0 \le x \le 1$ then $g(x)$____

(A) decreases in $\left[0, \frac{2}{3}\right]$ (B) decreases $\left[\frac{2}{3}, 1\right]$

(C) increases in $\left[0, \frac{2}{3}\right]$ (D) increases in $\left[\frac{2}{3}, 1\right]$

MMN039

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7. For the function $f(x) = x\cot^{-1}x, x \geq 0$

(A) there is atleast one $x \in (0, 1)$ for which $\cot^{-1}x = \frac{x}{1+x^2}$

(B) for atleast one x in the interval $(0, \infty)$, $f\left(x+\frac{2}{\pi}\right) - f(x) < 1$

(C) number of solution of the equation $f(x) = \sec x$ is 1

(D) $f'(x)$ is strictly decreasing in the interval $(0, \infty)$

MMN106

8. Which of the following inequalities are valid -

(A) $|\tan^{-1}x - \tan^{-1}y| \leq |x - y| \; \forall x, y \in R$ (B) $|\tan^{-1}x - \tan^{-1}y| \geq |x - y|$

(C) $|\sin x - \sin y| \leq |x - y|$ (D) $|\sin x - \sin y| \geq |x - y|$

MMN117

COMPREHENSION TYPE QUESTIONS

Paragraph for Question 9 to 11

If $f(x) = |x-2| + |x-4| + |x-6|$.

9. If the set of values of x for which $f(x)$ increases is $[a + 2, \infty)$, then 'a' is -

MMN046

10. If the set of values of x for which $f(x)$ decreases is $(-\infty, b - 1]$, then 'b' is -

MMN047

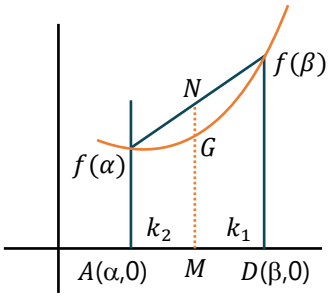
11. If $f(x)$ is symmetrical about the line $x = K$, then K is equal to –

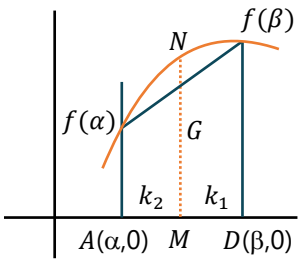
MMN048

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Paragraph for Question 12 to 14

For a double differentiable function $f(x)$ if $f''(x) \geq 0$ then $f(x)$ is concave upward and if $f''(x) \leq 0$ then $f(x)$ is concave downward





Here $M\left(\frac{k_1\alpha + k_2\beta}{k_1 + k_2}, 0\right)$

If $f(x)$ is a concave upward in $[a, b]$ and $\alpha, \beta \in [a, b]$ then $\frac{k_1 f(\alpha) + k_2 f(\beta)}{k_1 + k_2} \geq f\left(\frac{k_1\alpha + k_2\beta}{k_1 + k_2}\right)$,

where $k_1, k_2 \in R^+$

If $f(x)$ is a concave downward in $[a, b]$ and $\alpha, \beta \in [a, b]$ then $\frac{k_1 f(\alpha) + k_2 f(\beta)}{k_1 + k_2} \leq f\left(\frac{k_1\alpha + k_2\beta}{k_1 + k_2}\right)$,

where $k_1, k_2 \in R^+$

then answer the following:

For More Join: @IITJEE_Advanced

12. Which of the following is true

(A) $\frac{\sin\alpha+\sin\beta}{2} > \sin\left(\frac{\alpha+\beta}{2}\right)$; $\alpha, \beta \in (0, \pi)$ (B) $\frac{\sin\alpha+\sin\beta}{2} < \sin\left(\frac{\alpha+\beta}{2}\right)$; $\alpha, \beta \in (\pi, 2\pi)$

(C) $\frac{\sin\alpha+\sin\beta}{2} < \sin\left(\frac{\alpha+\beta}{2}\right)$; $\alpha, \beta \in (0, \pi)$ (D) None of these

MMN043

13. Which of the following is true

(A) $\frac{2^{\alpha}+2^{\beta+1}}{3} \le 2^{\frac{\alpha+2\beta}{3}}$

(B) $\frac{2\ell n\ \alpha+\ell n\beta}{3} \ge \ell n\left(\frac{2\alpha+\beta}{3}\right)$

(C) $\frac{\tan^{-1}\alpha+\tan^{-1}\beta}{2} \le \tan^{-1}\left(\frac{\alpha+\beta}{2}\right)$; $a, b \in R^-$

(D) $\frac{e^{\alpha}+2e^{\beta}}{3} \ge e^{\frac{\alpha+2\beta}{3}}$

MMN044

14. Let α, β and γ are three distinct real numbers and $f''(x) < 0$. Also $f(x)$ is increasing function and let $A = \frac{f^{-1}(\alpha)+f^{-1}(\beta)+f^{-1}(\gamma)}{3}$ and $B = f^{-1}\left(\frac{\alpha+\beta+\gamma}{3}\right)$, then order relation between A and B is ?

(A) $A > B$ (B) $A < B$ (C) $A = B$ (D) none of these

MMN045

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MATCHING LIST TYPE QUESTION

15. Match List-I with List-II and select the correct answer using the code given below the list.

	List - I		**List - II**
(P)	$f(x) = 2.\ e^{x^2-4x}$	(1)	Strictly increasing in $(2, \infty)$ & Strictly decreasing in $(-\infty, 2)$
(Q)	$f(x) = \frac{e^x}{x}$	(2)	Strictly increasing in $(0, 2)$ & Strictly decreasing in $(-\infty, 0) \cup (2, \infty)$
(R)	$f(x) = x^2\ e^{-x}$	(3)	Strictly increasing in $(1, \infty)$ & Strictly decreasing in $(-\infty, 0) \cup (0, 1)$
(S)	$f(x) = 2x^2 - ln\lvert x\rvert$	(4)	Strictly increasing in $\left(-\frac{1}{2}, 0\right) \cup \left(\frac{1}{2}, \infty\right)$ & Strictly decreasing in $\left(-\infty, -\frac{1}{2}\right) \cup \left(0, \frac{1}{2}\right)$

(A) P $\to$ 3; Q $\to$ 2; R $\to$ 4; S $\to$ 1 (B) P $\to$ 1; Q $\to$ 3; R $\to$ 2; S $\to$ 4

(C) P $\to$ 2; Q $\to$ 3; R $\to$ 4; S $\to$ 1 (D) P $\to$ 4; Q $\to$ 1; R $\to$ 3; S $\to$ 2

MMN049

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EXERCISE (S)

1. If $f(x) = 2e^x - ae^{-x} + (2a+1)x - 3$ increases for every $x \in R$ then the range of values of 'a' is $[k, \infty)$ find k.

MMN016

2. Find the sum of value of a, m and b if the function $f(x) = \begin{cases} 3 & x=0 \\ -x^2+3x+a & 0<x<1 \\ mx+b & 1 \le x \le 2 \end{cases}$ satisfy the hypothesis of the mean value theorem for the interval [0, 2].

MMN017

3. Find the least integral value of 'a' for which the function $f(x) = \sin x - a\sin 2x - \frac{1}{3}\sin 3x + 2ax$ increases throughout the number line.

MMN018

4. Let $f(x)$ be a strictly increasing function defined on $(0, \infty)$. If $f(2a^2 + a + 1) > f(3a^2 - 4a + 1)$. The range of a is $\left(0, \frac{1}{p}\right) \cup (q, r)$. Find $(p + q + r)$.

MMN019

5. If f and g are differentiable functions in [0, 1] satisfying $f(0) = 2 = g(1)$, $g(0) = 0$ and $f(1) = 6$, then $a.f'(c) = b.g'(c)$ for some $c \in (0,1)$, then the value of $\frac{b}{a}$ is :

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MMN020

6. If the sum of greatest & the least values of the function $f(x) = 12x^{\frac{4}{3}} - 6x^{\frac{1}{3}}$ in the interval $x \in [-1, 1]$, if they exist, is S then the value of $4S$ is

MMN050

7. Find the number of integral values of 'a' for which the function,

$f(x) = \left(1 - \frac{\sqrt{21 - 4a - a^2}}{a+1}\right)x^3 + 5x + \sqrt{7}$ is strictly increasing at every point of its domain.

MMN051

8. Find the minimum value of the function $f(x) = x^{3/2} + x^{-3/2} - 4\left(x + \frac{1}{x}\right)$ for all permissible real x.

MMN052

9. If $b > a$ and the minimum value of $|(x-a)^3| + |(x-b)^3|$, $x \in R$, is $\frac{(b-a)^p}{k}$, then the value of $(k - p)$ is

MMN053

10. If the set of values of x for which the inequality $\ln(1 + x) > x/(1 + x)$ is valid is $(a, b) \cup (b, \infty)$, then the value of $(2a - b)$ is

MMN054

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. If the function f given by $f(x) = x^3 - 3(a-2)x^2 + 3ax + 7$, for some $a \in R$ is increasing in (0, 1] and decreasing in [1, 5), then a root of the equation, $\frac{f(x)-14}{(x-1)^2} = 0 (x \neq 1)$ is : **[JEE (Main) 2019]**

(1) 6 (2) 5 (3) 7 (4) –7

MMN021

2. Let $f(x) = e^x - x$ and $g(x) = x^2 - x$, $\forall\ x \in$ R. Then the set of all $x \in$ R, where the function $h(x) = (fog)(x)$ is increasing, is : **[JEE (Main) 2019]**

(1) $\left[-1, \frac{-1}{2}\right] \cup \left[\frac{1}{2}, \infty\right)$ (2) $\left[0, \frac{1}{2}\right] \cup [1, \infty)$ (3) $\left[\frac{-1}{2}, 0\right] \cup [1, \infty)$ (4) $[0, \infty)$

MMN022

3. If m is the minimum value of k for which the function $f(x) = x\sqrt{kx - x^2}$ is increasing in the interval [0, 3] and M is the maximum value of f in [0, 3] when $k = m$, then the ordered pair (m, M) is equal to :

[JEE (Main) 2019]

(1) (4, $3\sqrt{2}$) (2) (4, $3\sqrt{3}$) (3) (3, $3\sqrt{3}$) (4) (5, $3\sqrt{6}$)

MMN023

4. The value of c in the Lagrange's mean value theorem for the function $f(x) = x^3 - 4x^2 + 8x + 11$, when $x \in [0, 1]$ is : **[JEE (Main) 2020]**

(1) $\frac{2}{3}$ (2) $\frac{\sqrt{7}-2}{3}$ (3) $\frac{4-\sqrt{5}}{3}$ (4) $\frac{4-\sqrt{7}}{3}$

MMN024

5. Let the function, $f : [-7, 0] \to R$ be continuous on $[-7, 0]$ and differentiable on $(-7, 0)$. If $f(-7) = -3$ and $f'(x) \leq 2$, for all $x \in (-7, 0)$, then for all such functions f, $f(-1) + f(0)$ lies in the interval : **[JEE (Main) 2020]**

(1) [–6, 20] (2) (–∞, 20] (3) (– ∞, 11] (4) [–3, 11]

MMN025

6. Let $f(x) = x\cos^{-1}(-\sin|x|)$, $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, then which of the following is true ? **[JEE (Main) 2020]**

(1) f' is decreasing in $\left(-\frac{\pi}{2}, 0\right)$ and increasing in $\left(0, \frac{\pi}{2}\right)$

(2) f is not differentiable at $x = 0$

(3) $f'(0) = -\frac{\pi}{2}$

(4) f' is increasing in $\left(-\frac{\pi}{2}, 0\right)$ and decreasing in $\left(0, \frac{\pi}{2}\right)$

MMN026

For More Join: @IITJEE_Advanced

7. Let f be any function continuous on $[a,b]$ and twice differentiable on (a,b). If for all $x \in (a,\ b), f'(x) > 0$ and $f''(x) < 0$, then for any $c \in (a,b)$, $\frac{f(c)-f(a)}{f(b)-f(c)}$ is greater than:

[JEE (Main) 2020]

(1) $\frac{b+a}{b-a}$ (2) $\frac{b-c}{c-a}$ (3) $\frac{c-a}{b-c}$ (4) 1

MMN027

8. Let $f : \mathbf{R} \to \mathbf{R}$ be defined as,

$$f(x)=\begin{cases}-55\,x, & \text{if } x<-5\\ 2x^3-3x^2-120x, & \text{if } -5\le x\le 4\\ 2x^3-3x^2-36x-336, & \text{if } x>4,\end{cases}$$

Let $A = \{x \in R : \text{f is increasing}\}$. Then A is equal to :

[JEE (Main) 2021]

(1) $(-\infty,-5)\cup(4,\infty)$ (2) $(-5,\infty)$

(3) $(-\infty,-5)\cup(-4,\infty)$ (4) $(-5,-4)\cup(4,\infty)$

MMN028

9. If Rolle's theorem holds for the function $f(x) = x^3 - ax^2 + bx - 4, x \in [1, 2]$ with $f'\left(\frac{4}{3}\right)=0$, then ordered pair (a, b) is equal to :

[JEE (Main) 2021]

(1) (5, 8) (2) (–5, 8) (3) (5, –8) (4) (–5, –8)

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MMN029

10. Consider the function $f : R \to R$ defined by $f(x)=\begin{cases}\left(2-\sin\left(\frac{1}{x}\right)\right)|x|, & x\neq 0\\ 0, & x=0\end{cases}$. Then f is :

(1) monotonic on $(-\infty, 0)\cup(0,\infty)$ (2) not monotonic on $(-\infty, 0)$ and $(0,\infty)$

(3) monotonic on $(0,\infty)$ only (4) monotonic on $(-\infty, 0)$ only

[JEE (Main) 2021]

MMN030

11. Let $f : R \to R$ and $g : R \to$ R be two functions defined by $f(x) = \log_e(x^2 + 1) - e^{-x} + 1$ and $g(x) = \frac{1-2e^{2x}}{e^x}$. Then, for which of the following range of α, the inequality

$f\left(g\left(\frac{(\alpha-1)^2}{3}\right)\right) > f\left(g\left(\alpha-\frac{5}{3}\right)\right)$ holds ?

[JEE (Main) 2022]

(1) (2, 3) (2) (–2, –1) (3) (1, 2) (4) (–1, 1)

MMN031

12. The number of real solutions of $x^7 + 5x^3 + 3x + 1 = 0$ is equal to _____. **[JEE (Main) 2022]**

(1) 0 (2) 1 (3) 3 (4) 5

MMN032

For More Join: @IITJEE_Advanced

13. The function $f(x)=xe^{x(1-x)}, x\in R,$ is **[JEE (Main) 2022]**

(1) increasing in $\left(-\frac{1}{2},1\right)$ (2) decreasing in $\left(\frac{1}{2},2\right)$

(3) increasing in $\left(-1,-\frac{1}{2}\right)$ (4) decreasing in $\left(-\frac{1}{2},\frac{1}{2}\right)$

MMN033

14. Let $f(x) = 2x + \tan^{-1}x$ and $g(x) = \log_e(\sqrt{1+x^2}+x)$, $x \in [0, 3]$. Then **[JEE (Main) 2023]**

(1) There exists $x\in[0,3]$ such that $f'(x)<g'(x)$

(2) $\max f(x) > \max g(x)$

(3) There exist $0 < x_1 < x_2 < 3$ such that $f(x) < g(x)$, $\forall x\in(x_1,x_2)$

(4) $\min f'(x) = 1 + \max g'(x)$

MMN034

15. Let $f:(0,1)\to\mathbb{R}$ be a function defined by

$f(x)=\frac{1}{1-e^{-x}}$, and $g(x)=(f(-x)-f(x))$. Consider two statements

(I) g is an increasing function in (0, 1)

(II) g is one-one in (0, 1)

Then, **[JEE (Main) 2023]**

(1) Only (I) is true (2) Only (II) is true

(3) Neither (I) nor (II) is true (4) Both (I) and (II) are true

MMN035

16. Let $g(x)=f(x)+f(1-x)$ and $f''(x)>0, x\in(0,1)$. If g is decreasing in the interval $(0, \alpha)$ and increasing in the interval $(\alpha, 1)$, then $\tan^{1}(2\alpha)+\tan^{-1}\left(\frac{1}{\alpha}\right)+\tan^{-1}\left(\frac{\alpha+1}{\alpha}\right)$ is equal to :

[JEE (Main) 2023]

(1) $\frac{3\pi}{2}$ (2) π (3) $\frac{5\pi}{4}$ (4) $\frac{3\pi}{4}$

MMN036

17. Let $f:\mathbb{R}\to\mathbb{R}$ be a thrice differentiable function such that $f(0) = 0, f(1) = 1, f(2) = -1, f(3) = 2$ and $f(4) = -2$. Then, the minimum number of zeros of $(3f'f'' + ff''')(x)$ is .....

[JEE (Main) 2024]

MMN118

18. For the function $f(x) = \sin x + 3x - \frac{2}{\pi}(x^2 + x)$, where $x \in \left[0,\frac{\pi}{2}\right]$, consider the following two statements :

(I) f is increasing in $\left(0,\frac{\pi}{2}\right)$.

(II) f' is decreasing in $\left(0,\frac{\pi}{2}\right)$.

Between the above two statements, **[JEE (Main) 2024]**

(1) only (I) is true. (2) only (II) is true.

(3) neither (I) nor (II) is true. (4) both (I) and (II) are true.

MMN119

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19. Let $g(x)=3f\left(\frac{x}{3}\right)+f(3-x)$ and $f''(x)>0$ for all $x\in(0,3)$. If g is decreasing in $(0, \alpha)$ and increasing in $(\alpha, 3)$, then 8α is **[JEE (Main) 2024]**

(1) 24 (2) 0 (3) 18 (4) 20

MMN120

20. Consider the function $f:\left[\frac{1}{2},1\right]\to R$ defined by $f(x)=4\sqrt{2}x^3-3\sqrt{2}x-1$. Consider the statements

(I) The curve $y = f(x)$ intersects the x-axis exactly at one point

(II) The curve $y = f(x)$ intersects the x-axis at $x=\cos\frac{\pi}{12}$

Then **[JEE (Main) 2024]**

(1) Only (II) is correct (2) Both (I) and (II) are incorrect

(3) Only (I) is correct (4) Both (I) and (II) are correct

MMN121

21. Let $g : R\to R$ be a non constant twice differentiable such that $g'\left(\frac{1}{2}\right)=g'\left(\frac{3}{2}\right)$. If a real valued function f is defined as $f(x)=\frac{1}{2}\left[g(x)+g(2-x)\right]$, then

[JEE (Main) 2024]

(1) $f''(x) = 0$ for atleast two x in $(0, 2)$

(2) $f''(x) = 0$ for exactly one x in $(0, 1)$

(3) $f''(x) = 0$ for no x in $(0, 1)$

(4) $f'\left(\frac{3}{2}\right)+f'\left(\frac{1}{2}\right)=1$

MMN122

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EXERCISE - JEE (Advanced) PYQ

1. Let $f:(0,1)\to R$ be defined by $f(x)=\frac{b-x}{1-bx}$, where b is a constant such that $0<b<1$. Then **[JEE (Advanced) 2011]**

(A) f is not invertible on $(0,1)$ (B) $f\neq f^{-1}$ on $(0,1)$ and $f'(b)=\frac{1}{f'(0)}$

(C) $f=f^{-1}$ on $(0,1)$ and $f'(b)=\frac{1}{f'(0)}$ (D) f^{-1} is differentiable on $(0,1)$

MMN055

2. The number of distinct real roots of $x^4-4x^3+12x^2+x-1=0$ is **[JEE (Advanced) 2011]**

MMN056

3. The number of points in $(-\infty,\infty)$, for which $x^2-x\sin x-\cos x=0$, is **[JEE (Advanced) 2013]**

(A) 6 (B) 4 (C) 2 (D) 0

MMN057

4. Let $f(x)=x\sin\pi x$, $x>0$. Then for all natural numbers n, $f'(x)$ vanishes at - **[JEE (Advanced) 2013]**

(A) a unique point in the interval $\left(n,n+\frac{1}{2}\right)$ (B) a unique point in the interval $\left(n+\frac{1}{2},n+1\right)$

(C) a unique point in the interval $(n,n+1)$ (D) two points in the interval $(n,n+1)$

MMN058

5. Let $a\in\mathbb{R}$ and let $f:\mathbb{R}\to\mathbb{R}$ be given by $f(x)=x^5-5x+a$. Then **[JEE (Advanced) 2014]**

(A) $f(x)$ has three real roots if $a>4$ (B) $f(x)$ has only one real roots if $a>4$

(C) $f(x)$ has three real roots if $a<-4$ (D) $f(x)$ has three real roots if $-4<a<4$

MMN059

6. Let $f:\left(-\frac{\pi}{2},\frac{\pi}{2}\right)\to\mathbb{R}$ be given by $f(x)=(\log(\sec x+\tan x))^3$ Then :-

[JEE (Advanced) 2014]

(A) $f(x)$ is an odd function (B) $f(x)$ is a one-one function

(C) $f(x)$ is an onto function (D) $f(x)$ is an even function

MMN060

7. Let $f, g:[-1,2]\to\mathbb{R}$ be continuous function which are twice differentiable on the interval $(-1,2)$. Let the values of f and g at the points -1, 0 and 2 be as given in the following table :

	$x=-1$	$x=0$	$x=2$
$f(x)$	3	6	0
$g(x)$	0	1	-1

In each of the intervals $(-1,0)$ and $(0,2)$ the function $(f-3g)''$ never vanishes. Then the correct statement(s) is(are) **[JEE (Advanced) 2015]**

(A) $f'(x)-3g'(x)=0$ has exactly three solutions in $(-1,0)\cup(0,2)$

(B) $f'(x)-3g'(x)=0$ has exactly one solution in $(-1,0)$

(C) $f'(x)-3g'(x)=0$ has exactly one solutions in $(0,2)$

(D) $f'(x)-3g'(x)=0$ has exactly two solutions in $(-1,0)$ and exactly two solutions in $(0,2)$

MMN061

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Answer Q.8, Q.9 and Q.10 by appropriately matching the information given in the three columns of the following table. **[JEE (Advanced) 2017]**

Let $f(x) = x + \log_e x - x \log_e x$, $x \in (0, \infty)$.

* Column 1 contains information about zeros of $f(x)$, $f'(x)$ and $f''(x)$.
* Column 2 contains information about the limiting behavior of $f(x)$, $f'(x)$ and $f''(x)$ at infinity.
* Column 3 contains information about increasing/decreasing nature of $f(x)$ and $f'(x)$.

Column 1		**Column 2**		**Column 3**	
(I)	$f(x) = 0$ for some $x \in (1, e^2)$	(i)	$\lim_{x\to\infty} f(x) = 0$	(P)	f is increasing in (0,1)
(II)	$f'(x) = 0$ for some $x \in (1, e)$	(ii)	$\lim_{x\to\infty} f(x) = -\infty$	(Q)	f is decreasing in (e, e^2)
(III)	$f'(x) = 0$ for some $x \in (0, 1)$	(iii)	$\lim_{x\to\infty} f'(x) = -\infty$	(R)	f' is increasing in (0,1)
(IV)	$f''(x) = 0$ for some $x \in (1, e)$	(iv)	$\lim_{x\to\infty} f''(x) = 0$	(S)	f' is decreasing in (e, e^2)

8. Which of the following options is the only **CORRECT** combination ?

(A) (IV) (i) (S) (B) (I) (ii) (R) (C) (III) (iv) (P) (D) (II) (iii) (S)

MMN062

9. Which of the following options is the only **CORRECT** combination ?

(A) (III) (iii) (R) (B) (I) (i) (P) (C) (IV) (iv) (S) (D) (II) (ii) (Q)

MMN063

10. Which of the following options is the only **INCORRECT** combination ?

(A) (II) (iii) (P) (B) (II) (iv) (Q) (C) (I) (iii) (P) (D) (III) (i) (R)

MMN064

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11. If $f : \mathbb{R} \to \mathbb{R}$ is a twice differentiable function such that $f''(x) > 0$ for all $x \in \mathbb{R}$, and $f\left(\frac{1}{2}\right) = \frac{1}{2}$, $f(1) = 1$, then **[JEE (Advanced) 2017]**

(A) $0 < f'(1) \le \frac{1}{2}$ (B) $f'(1) \le 0$ (C) $f'(1) > 1$ (D) $\frac{1}{2} < f'(1) \le 1$

MMN065

12. For a polynomial $g(x)$ with real coefficient, let m_g denote the number of distinct real roots of $g(x)$. Suppose S is the set of polynomials with real coefficients defined by

$S = \{(x^2 - 1)^2 (a_0 + a_1 x + a_2 x^2 + a_3 x^3) : a_0, a_1, a_2, a_3 \in R\}$.

For a polynomial f, let f' and f'' denote its first and second order derivatives, respectively. Then the minimum possible value of $(m_{f'} + m_{f''})$, where $f \in S$, is ___ **[JEE (Advanced) 2020]**

MMN066

13. $f : R \to R$ is defined by $f(x) = |x|(x - \sin x)$, then which of the following statements is **TRUE** ? **[JEE (Advanced) 2020]**

(A) f is one-one, but **NOT** onto
(B) f is onto, but **NOT** one-one
(C) f is **BOTH** one-one and onto
(D) f is **NEITHER** one-one **NOR** onto

MMN067

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14. Let $f:\mathbb{R}\to\mathbb{R}$ be defined by

$$f(x)=\frac{x^2-3x-6}{x^2+2x+4}.$$

Then which of the following statements is (are) TRUE ? **[JEE (Advanced) 2021]**

(A) f is decreasing in the interval (–2, –1)

(B) f is increasing in the interval (1, 2)

(C) f is onto

(D) Range of f is $\left[-\frac{3}{2},2\right]$

MMN068

15. Let S be the set of all twice differentiable functions f from $\mathbb{R}$ to $\mathbb{R}$ such that $\frac{d^2f}{dx^2}(x)>0$ for all $x\in(-1,1)$. For $f\in S$, let X_f be the number of points $x\in(-1,1)$ for which $f(x)=x$. Then which of the following statements is(are) true? **[JEE (Advanced) 2023]**

(A) There exists a function $f\in S$ such that $X_f=0$

(B) For every function $f\in S$, we have $X_f\le 2$

(C) There exists a function $f\in S$ such that $X_f=2$

(D) There does **NOT** exist any function f in S such that $X_f=1$

MMN123

16. Let the function $f:\mathbb{R}\to\mathbb{R}$ be defined by

$f(x)=\frac{\sin x}{e^{\pi x}}\frac{(x^{2023}+2024x+2025)}{(x^2-x+3)}+\frac{2}{e^{\pi x}}\frac{(x^{2023}+2024x+2025)}{(x^2-x+3)}$. Then the number of solutions of $f(x)=0$ in $\mathbb{R}$ is _____. **[JEE (Advanced) 2024]**

MMN124

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	A	C	B	C	A	C	A	C	B
Que.	11	12	13	14	15					
Ans.	A	A	B	A	B					

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B	A,D	A,C,D	A,B,C	A,D	B,C	B,D	A,C	2	5
Que.	11	12	13	14	15					
Ans.	4	C	D	A	B					

EXERCISE (S)

1. 0 **2.** 8 **3.** 1 **4.** 9 **5.** 3

6. 63 **7.** 8 **8.** –10 **9.** 1 **10.** –2

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	2	2	4	2	1	3	4	1	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	2	1	2	4	2	5	4	3	4
Que.	21									
Ans.	1									

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	2	C	B,C	B,D	A,B,C	B,C	D	D	D
Que.	11	12	13	14	15	16				
Ans.	C	5	C	A,B	A,B,C	1				

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